Multivariate Calculus Lecture Notes
Section 12.7 – Cylindrical and Spherical Coordinates

I. Cylindrical coordinates
   a. Extension of the polar coordinate system to three dimensions
   b. A point $P$ is represented by $(r, \theta, z)$. (Do example in space, in the classroom.)
   c. Explanation of coordinates
      i. $(r, \theta)$ are the polar coordinates of the projection of $P$ onto the $xy$-plane.
      ii. $r$ is the directed distance from the origin to the projection.
      iii. $\theta$ is the angle between the projection and the positive $x$-axis.
      iv. $z$ is the directed distance to the point from the $xy$-plane.
   d. Graphically:

   e. Useful for surfaces that have symmetry about the $z$-axis
   f. Conversions
      i. Cylindrical to rectangular
         1. $x = r\cos\theta$
         2. $y = r\sin\theta$
         3. $z = z$
      ii. Rectangular to cylindrical
         1. $r^2 = x^2 + y^2$
         2. $\tan\theta = y/x$
         3. $z = z$
   g. Cylindrical coordinates are not unique!
      i. For example, $(1, \pi/4, 0)$ and $(-1, 5\pi/4, 0)$ denote the same point in three-space.
      ii. The same sort of thing is true with polar coordinates.
   h. Examples
      i. $r = 4$ is a cylinder: $r^2 = 16 \implies x^2 + y^2 = 16$
      ii. $\theta = \pi/4$ is a vertical plane containing the $z$-axis: $\tan\theta = \tan(\pi/4) = 1 \implies y/x = 1 \implies y = x$
      iii. $z = 3$ is a horizontal plane, same as with rectangular coordinates
      iv. $r = z$ is a cone: $r^2 = z^2 \implies x^2 + y^2 = z^2$
      v. $r = \sqrt{z}$ is a paraboloid: $r^2 = z \implies x^2 + y^2 = z$
      vi. $r = \theta$ is a spiral with infinitely high walls

II. Spherical coordinates
   a. A point $P$ in space is represented by $(\rho, \theta, \phi)$, where
      i. $\rho$ is the distance between $P$ and the origin; $\rho \geq 0$.
      ii. $\theta$ is the polar angle – the same as with cylindrical coordinates.
      iii. $\phi$ is the *angle of declination* – the angle between the positive $z$-axis and the line segment $\overline{OP}$; $0 \leq \phi \leq \pi$.
   b. Graphically:

   c. Useful for surfaces that have symmetry about a point

d. Conversions
   i. Spherical to rectangular
      1. $z/\rho = \cos\phi \implies z = \rho\cos\phi$
      2. $r/\rho = \sin\phi \implies r = \rho\sin\phi$
      3. $x = r\cos\theta = \rho\sin\phi\cos\theta$
      4. $y = r\sin\theta = \rho\sin\phi\sin\theta$
   ii. Rectangular to spherical
      1. $\rho^2 = x^2 + y^2 + z^2$
      2. $\phi = \arccos(z/\sqrt{x^2+y^2+z^2})$
      3. $\tan\theta = y/x$

e. Examples
   i. $\rho = 4$ is a sphere of radius 4
   ii. $\theta = \pi/4$ is now a vertical half-plane containing the $z$-axis
   iii. $\phi = \pi/4$ is a cone

III. Additional examples

a. Sketch the solids described by the following equations.
   i. $0 \le \theta \le \pi/2$, $0 \le r \le 2$, $0 \le z \le 4$

   ii. $0 \le \theta \le 2\pi$, $0 \le r \le 2$, $r \le z \le 2$

   iii. $0 \le \theta \le 2\pi$, $0 \le \rho \le 1$, $0 \le \phi \le \pi/6$

b. Describe the following solids by inequalities in cylindrical or spherical coordinates.
   i. A cylindrical shell 8 meters long with an inside radius of 0.75 meters and an outside radius of 1.25 meters
      - $0.75 \le r \le 1.25$, $0 \le z \le 8$, $0 \le \theta \le 2\pi$
   ii. The solid that remains after a hole 1 inch in radius is drilled through the center of a sphere 6 with a 6-inch radius.
      1. The equation for the cylinder is $r = 1$, and the equation for the sphere is $\rho = 6$.
      2. Cylindrical
         - We have $\rho = 6 \implies x^2 + y^2 + z^2 = 36$.
         - Then $z = \pm\sqrt{36 - r^2}$ (top and bottom halves).
         - The answer is $0 \le \theta \le 2\pi$, $1 \le r \le 6$, $-\sqrt{36-r^2} \le z \le \sqrt{36-r^2}$.
      3. Spherical
         - We have $r = 1 \implies \rho\sin\phi = 1 \implies \rho = \csc\phi$.
         - The bounds on $\phi$ must be determined graphically:

         - So we have $0 \le \theta \le 2\pi$, $\arcsin(1/6) \le \phi \le \pi - \arcsin(1/6)$, $\csc\phi \le \rho \le 6$.

**Homework**: Section 12.7 #7, 14, 20, 24, 40, 45, 52, 56, 59, 76.